



FATE MONTHLY REPORT

Doc.No : FATE-001
Version : 1.0
Date : 13-11-2023



FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
October 2023



PO102958/ESO/20/95952/FLAB

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1 Title for Night 1

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1.1 Title for Figures of Night 1

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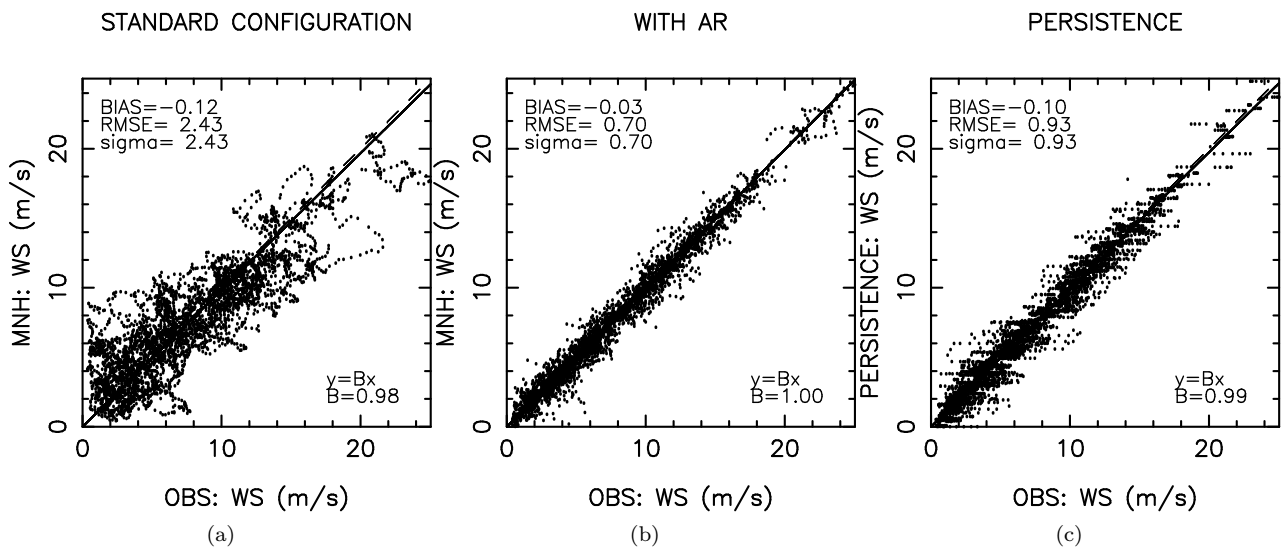


Figure 1: wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

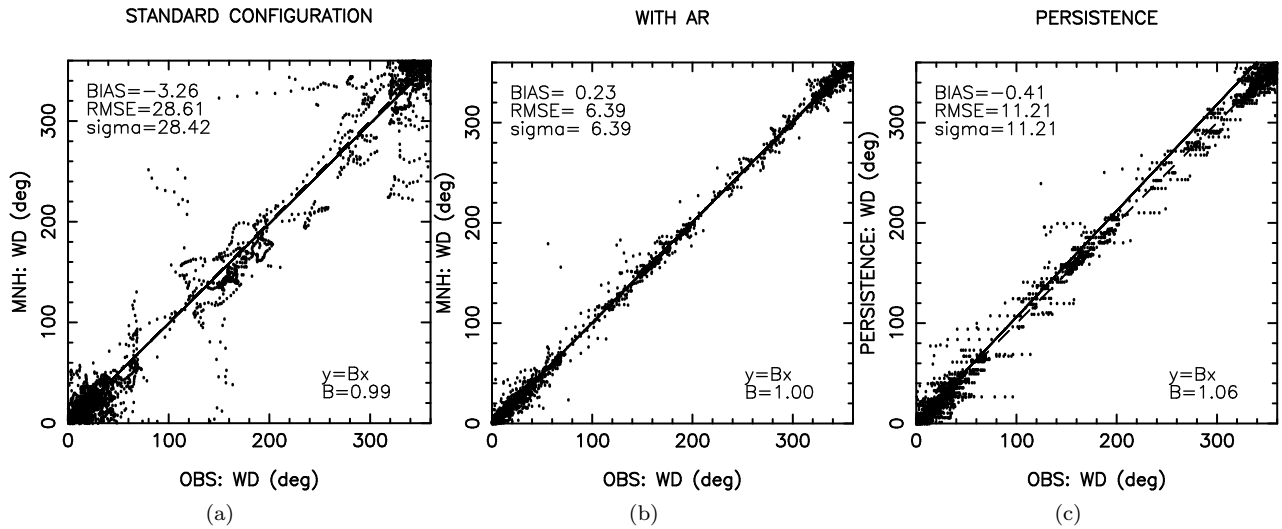


Figure 2: wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

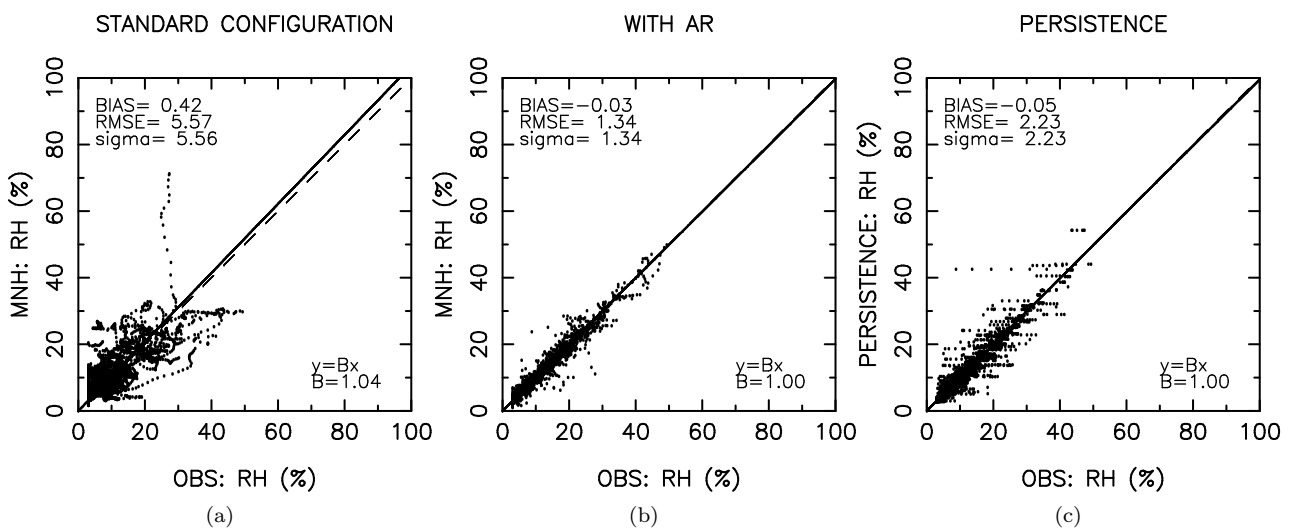


Figure 3: relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

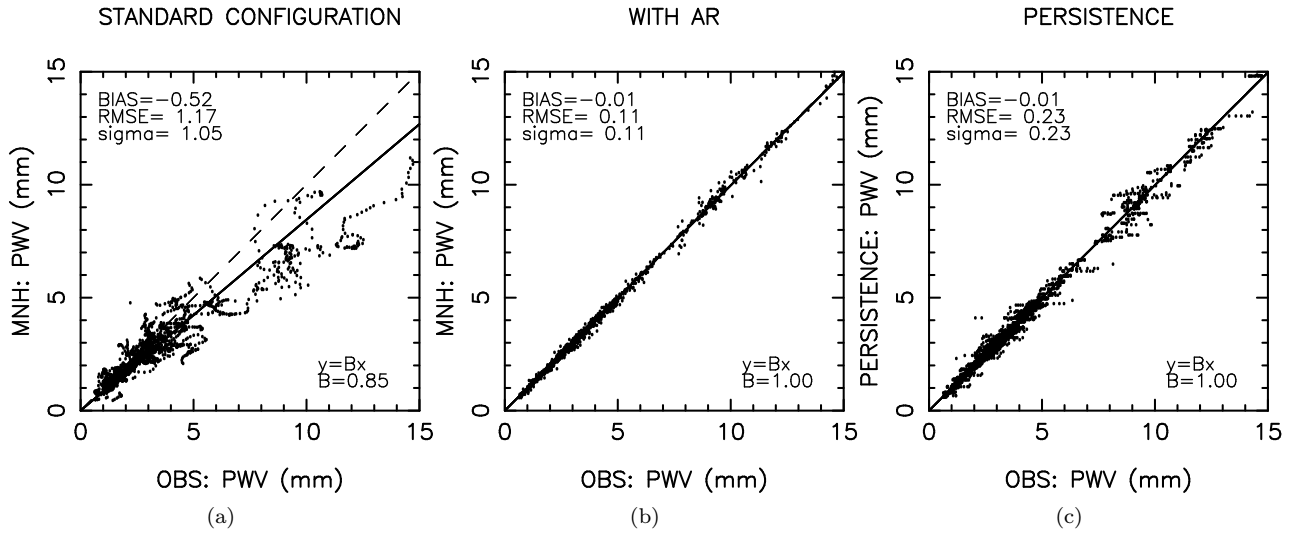


Figure 4: precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

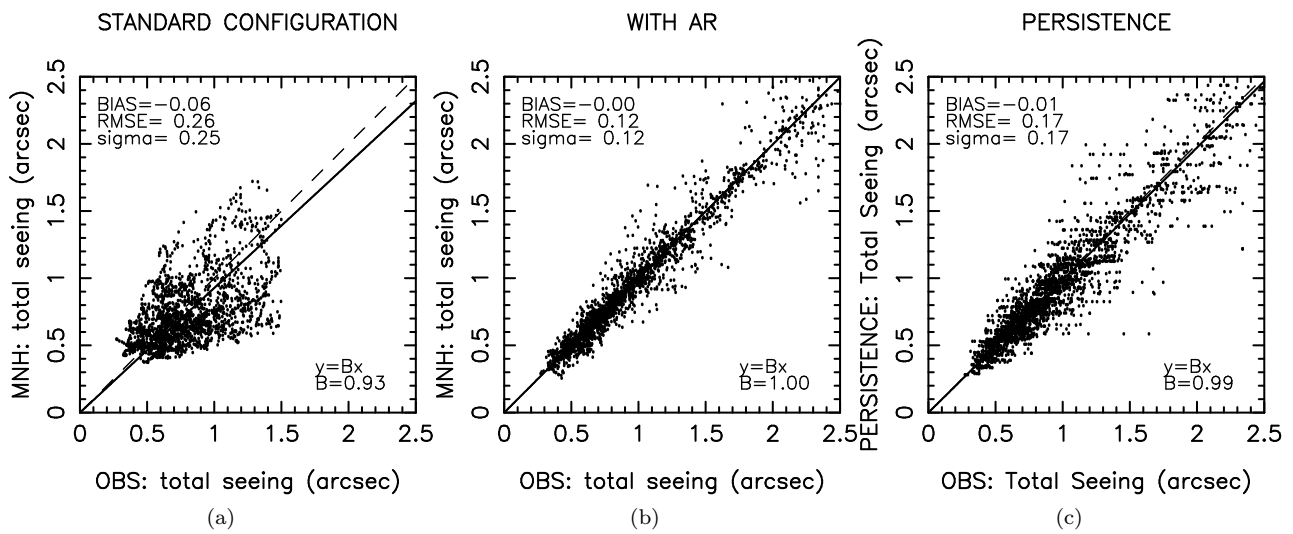


Figure 5: total seeing (arcsec): (a) STANDARD CONFIGURATION ($< 1.5''$), (b) WITH AR (1H), (c) PERSISTENCE (1H).

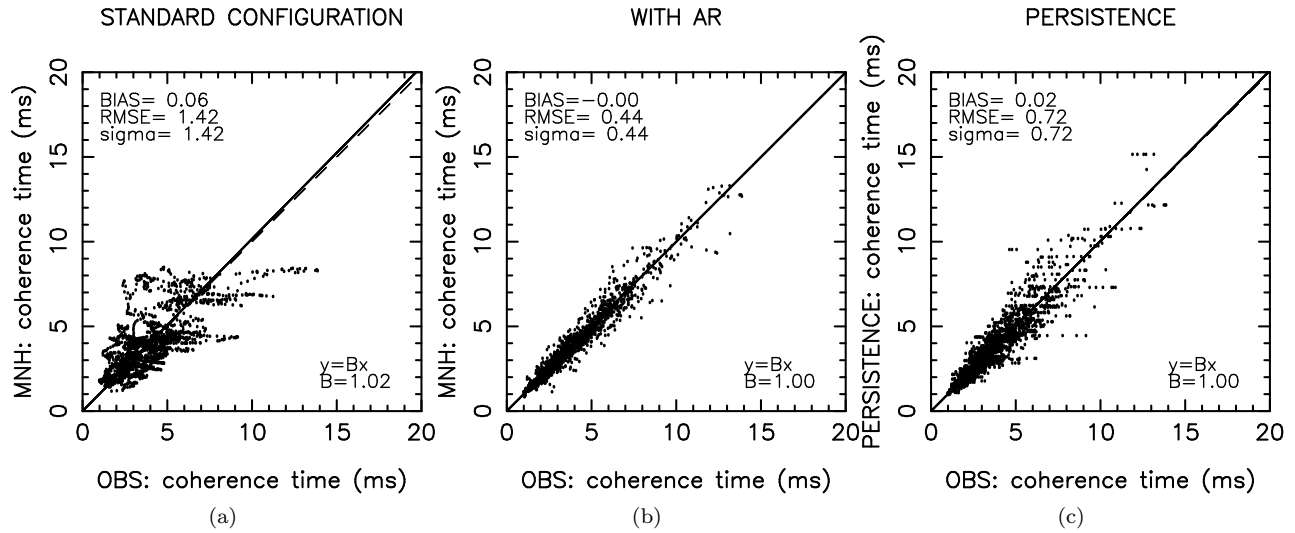


Figure 6: coherence time (ms): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

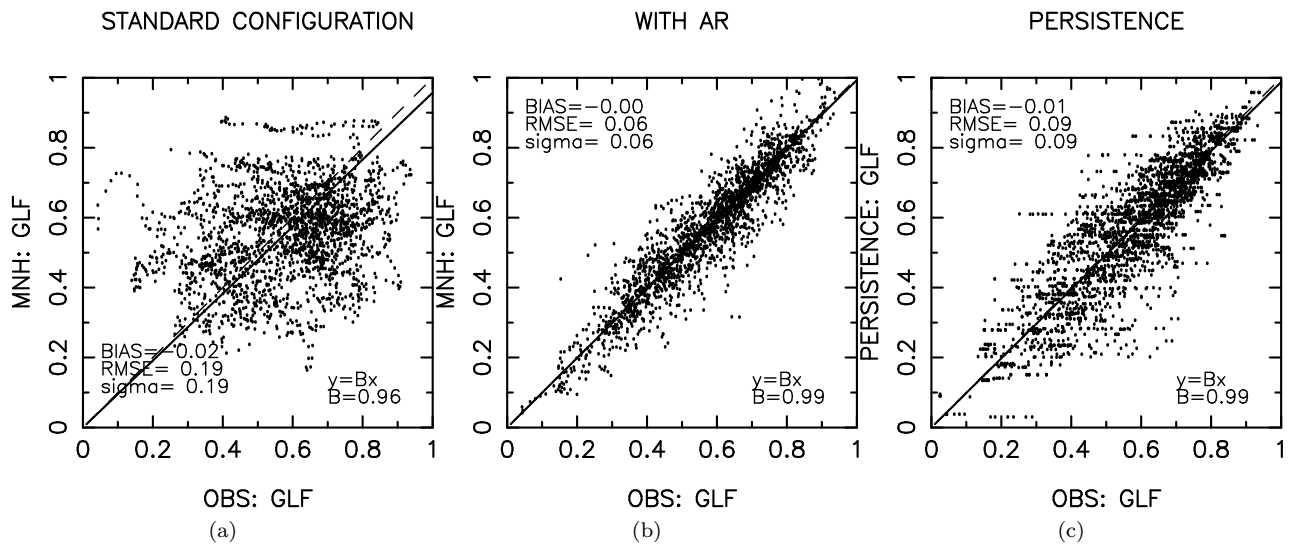


Figure 7: ground layer fraction (dimensionless): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

1.2 Title for Statistics of Night 1

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VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS (m·s ⁻¹)	-0.12	2.43	2.43	-0.76	2.25	2.12
WD (degree)	-3.26	28.61	28.42	-2.14	26.84	26.75
RH (percent)	0.42	5.57	5.56	0.03	6.13	6.13
PWV (mm)	-0.52	1.17	1.05	-0.75	1.51	1.32
SEE (arcsec)	-0.06	0.26	0.25	-0.01	0.24	0.24
TAU (ms)	0.06	1.42	1.42	-0.32	1.32	1.28
GLF (dimensionless)	-0.02	0.19	0.19	-0.09	0.19	0.16

Table 1: Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (October-2023)

FATE - Monthly Report

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.04	0.68	0.68		0.91	0.90
WD (degree)	0.11	6.59	6.58		11.52	11.46
RH (percent)	-0.03	1.54	1.54		2.75	2.75
PWV (mm)	-0.01	0.14	0.14		0.27	0.27
SEE (arcsec)	0.01	0.12	0.12		0.18	0.18
TAU (ms)	-0.01	0.48	0.48		0.81	0.81
GLF (dimensionless)	-0.01	0.06	0.06		0.09	0.09

Table 2: Last month (October-2023): statistics for variables with AR (1H) *vs* persistence (1H)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.03	0.70	0.70		0.93	0.93
WD (degree)	0.23	6.39	6.39		11.21	11.21
RH (percent)	-0.03	1.34	1.34		2.23	2.23
PWV (mm)	-0.01	0.11	0.11		0.23	0.23
SEE (arcsec)	-0.00	0.12	0.12		0.17	0.17
TAU (ms)	-0.00	0.44	0.44		0.72	0.72
GLF (dimensionless)	-0.00	0.06	0.06		0.09	0.09

Table 3: Incremental month (i.e., since the beginning of service): statistics for variables with AR (1H) *vs* persistence (1H)

1.3 Title for Contingency tables of Night 1

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wind speed ms^{-1}	Observations		
	ws < 4.9	4.9 < ws < 9.8	ws > 9.8
Model data	ws < 4.9	917	203
	4.9 < ws < 9.8	444	1003
	ws > 9.8	1	156
Sample size: 6588; PC=72.3%; EBD=0.0%; POD1=67.3%; POD2=73.6%; POD3=76.0%			

Table 4: Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.9	4.9 < ws < 9.8	ws > 9.8
Model data	ws < 4.9	1308	64
	4.9 < ws < 9.8	54	1252
	ws > 9.8	0	46
Sample size: 6588; PC=94.3%; EBD=0.0%; POD1=96.0%; POD2=91.9%; POD3=95.0%			

Table 5: Contingency table for variable wind speed (ms^{-1}) processed with AR (1H)

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wind direction degree	Observations			
	North	East	South	West
Model data	North	2318	101	14
	East	49	133	38
	South	5	36	374
	West	99	17	1

Sample size: 6588; PC=86.5%; EBD=1.1%; POD1=93.8%; POD2=46.3%; POD3=87.6%; POD4=44.4%

Table 6: Contingency table for wind direction (degree) in standard configuration

wind direction degree	Observations			
	North	East	South	West
Model data	North	2452	30	0
	East	14	243	5
	South	0	14	418
	West	5	0	4

Sample size: 6588; PC=97.7%; EBD=0.0%; POD1=99.2%; POD2=84.7%; POD3=97.9%; POD4=97.0%

Table 7: Contingency table for wind direction (degree) in standard configuration

FATE - Monthly Report

relative humidity percent		Observations		
		rh < 7.7	7.7 < rh < 12.1	rh > 12.1
Model data	rh < 7.7	745	373	73
	7.7 < rh < 12.1	365	646	243
	rh > 12.1	212	304	1004
Sample size: 6588; PC=60.4%; EBD=7.2%; POD1=56.4%; POD2=48.8%; POD3=76.1%				

Table 8: Contingency table for variable relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.7	7.7 < rh < 12.1	rh > 12.1
Model data	rh < 7.7	1244	55	4
	7.7 < rh < 12.1	75	1222	58
	rh > 12.1	3	46	1258
Sample size: 6588; PC=93.9%; EBD=0.2%; POD1=94.1%; POD2=92.4%; POD3=95.3%				

Table 9: Contingency table for variable relative humidity (percent) processed with AR (1H)

precipitable water vapor mm		Observations		
		pwv < 2.0	2.0 < pwv < 3.3	pwv > 3.3
Model data	pwv < 2.0	630	128	0
	2.0 < pwv < 3.3	27	468	181
	pwv > 3.3	0	61	475
Sample size: 3294; PC=79.8%; EBD=0.0%; POD1=95.9%; POD2=71.2%; POD3=72.4%				

Table 10: Contingency table for variable precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.0	2.0 < pwv < 3.3	pwv > 3.3
Model data	pwv < 2.0	649	13	0
	2.0 < pwv < 3.3	8	634	12
	pwv > 3.3	0	10	644
Sample size: 3294; PC=97.8%; EBD=0.0%; POD1=98.8%; POD2=96.5%; POD3=98.2%				

Table 11: Contingency table for variable precipitable water vapor (mm) processed with AR (1H)

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total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.9	see > 0.9
Model data	see < 0.6	452	244	102
	0.6 < see < 0.9	162	290	296
	see > 0.9	43	123	258
Sample size: 5184; PC=50.8%; EBD=7.4%; POD1=68.8%; POD2=44.1%; POD3=39.3%				

Table 12: Contingency table for variable total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.7	0.7 < see < 1.0	see > 1.0
Model data	see < 0.7	425	44	1
	0.7 < see < 1.0	35	374	37
	see > 1.0	1	43	422
Sample size: 3024; PC=88.4%; EBD=0.1%; POD1=92.2%; POD2=81.1%; POD3=91.7%				

Table 13: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.7	0.7 < see < 1.0	see > 1.0
Model data	see < 0.7	425	44	1
	0.7 < see < 1.0	35	374	37
	see > 1.0	1	43	422
Sample size: 3024; PC=88.4%; EBD=0.1%; POD1=92.2%; POD2=81.1%; POD3=91.7%				

Table 14: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.7	0.7 < see < 1.0	see > 1.0
Model data	see < 0.7	360	39	1
	0.7 < see < 1.0	19	307	38
	see > 1.0	1	34	341
Sample size: 2484; PC=88.4%; EBD=0.2%; POD1=94.7%; POD2=80.8%; POD3=89.7%				

Table 15: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.9	see > 0.9
Model data	see < 0.6	692	72	0
	0.6 < see < 0.9	68	629	66
	see > 0.9	0	59	694
Sample size: 5184; PC=88.4%; EBD=0.0%; POD1=91.1%; POD2=82.8%; POD3=91.3%				

Table 16: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.9	see > 0.9
	see < 0.6	692	72	0
Model data	0.6 < see < 0.9	68	629	66
	see > 0.9	0	59	694
Sample size: 5184; PC=88.4%; EBD=0.0%; POD1=91.1%; POD2=82.8%; POD3=91.3%				

Table 17: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.9	see > 0.9
	see < 0.6	692	72	0
Model data	0.6 < see < 0.9	68	629	66
	see > 0.9	0	59	694
Sample size: 5184; PC=88.4%; EBD=0.0%; POD1=91.1%; POD2=82.8%; POD3=91.3%				

Table 18: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.9	see > 0.9
	see < 0.6	692	72	0
Model data	0.6 < see < 0.9	68	629	66
	see > 0.9	0	59	694
Sample size: 5184; PC=88.4%; EBD=0.0%; POD1=91.1%; POD2=82.8%; POD3=91.3%				

Table 19: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.9	see > 0.9
	see < 0.6	692	72	0
Model data	0.6 < see < 0.9	68	629	66
	see > 0.9	0	59	694
Sample size: 5184; PC=88.4%; EBD=0.0%; POD1=91.1%; POD2=82.8%; POD3=91.3%				

Table 20: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.9	see > 0.9
	see < 0.6	610	0	116
Model data	0.6 < see < 0.9	4	636	116
	see > 0.9	43	21	540
Sample size: 5184; PC=90.7%; EBD=8.1%; POD1=92.8%; POD2=96.8%; POD3=69.9%				

Table 21: Contingency table for variable total seeing (arcsec) in standard configuration and accuracy 0.24

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.7	0.7 < see < 1.0	see > 1.0
	see < 0.7	459	0	3
Model data	0.7 < see < 1.0	1	456	3
	see > 1.0	1	5	457
Sample size: 3024; PC=99.3%; EBD=0.3%; POD1=99.6%; POD2=98.9%; POD3=98.7%				

Table 22: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.7	0.7 < see < 1.0	see > 1.0
	see < 0.7	459	0	3
Model data	0.7 < see < 1.0	1	456	3
	see > 1.0	1	5	457
Sample size: 3024; PC=99.3%; EBD=0.3%; POD1=99.6%; POD2=98.9%; POD3=98.7%				

Table 23: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.7	0.7 < see < 1.0	see > 1.0
	see < 0.7	378	0	2
Model data	0.7 < see < 1.0	1	376	2
	see > 1.0	1	4	378
Sample size: 2484; PC=99.3%; EBD=0.3%; POD1=99.5%; POD2=98.9%; POD3=99.0%				

Table 24: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.9	see > 0.9
	see < 0.6	760	0	2
Model data	0.6 < see < 0.9	0	754	2
	see > 0.9	0	6	758
Sample size: 5184; PC=99.6%; EBD=0.1%; POD1=100.0%; POD2=99.2%; POD3=99.5%				

Table 25: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.9	see > 0.9
	see < 0.6	760	0	2
Model data	0.6 < see < 0.9	0	754	2
	see > 0.9	0	6	758
Sample size: 5184; PC=99.6%; EBD=0.1%; POD1=100.0%; POD2=99.2%; POD3=99.5%				

Table 26: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.9	see > 0.9
Model data	see < 0.6	760	0	2
	0.6 < see < 0.9	0	754	2
	see > 0.9	0	6	758
Sample size: 5184; PC=99.6%; EBD=0.1%; POD1=100.0%; POD2=99.2%; POD3=99.5%				

Table 27: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.9	see > 0.9
Model data	see < 0.6	760	0	2
	0.6 < see < 0.9	0	754	2
	see > 0.9	0	6	758
Sample size: 5184; PC=99.6%; EBD=0.1%; POD1=100.0%; POD2=99.2%; POD3=99.5%				

Table 28: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.9	see > 0.9
Model data	see < 0.6	760	0	2
	0.6 < see < 0.9	0	754	2
	see > 0.9	0	6	758
Sample size: 5184; PC=99.6%; EBD=0.1%; POD1=100.0%; POD2=99.2%; POD3=99.5%				

Table 29: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	420	169	11
	$2.8 < \tau < 4.2$	189	353	175
	$\tau > 4.2$	65	152	488
Sample size: 5184; PC=62.4%; EBD=3.8%; POD1=62.3%; POD2=52.4%; POD3=72.4%				

Table 30: Contingency table for variable coherence time (ms) in standard configuration and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.4$	$2.4 < \tau < 3.5$	$\tau > 3.5$
Model data	$\tau < 2.4$	392	28	0
	$2.4 < \tau < 3.5$	33	375	28
	$\tau > 3.5$	1	23	398
Sample size: 3024; PC=91.2%; EBD=0.1%; POD1=92.0%; POD2=88.0%; POD3=93.4%				

Table 31: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.4$	$2.4 < \tau < 3.5$	$\tau > 3.5$
Model data	$\tau < 2.4$	392	28	0
	$2.4 < \tau < 3.5$	33	375	28
	$\tau > 3.5$	1	23	398
Sample size: 3024; PC=91.2%; EBD=0.1%; POD1=92.0%; POD2=88.0%; POD3=93.4%				

Table 32: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.4$	$2.4 < \tau < 3.6$	$\tau > 3.6$
Model data	$\tau < 2.4$	319	25	0
	$2.4 < \tau < 3.6$	34	298	22
	$\tau > 3.6$	1	30	331
Sample size: 2484; PC=89.4%; EBD=0.1%; POD1=90.1%; POD2=84.4%; POD3=93.8%				

Table 33: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	618	54	0
	$2.8 < \tau < 4.2$	56	561	44
	$\tau > 4.2$	0	59	630
Sample size: 5184; PC=89.5%; EBD=0.0%; POD1=91.7%; POD2=83.2%; POD3=93.5%				

Table 34: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

FATE - Monthly Report

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	618	54	0
	$2.8 < \tau < 4.2$	56	561	44
	$\tau > 4.2$	0	59	630
Sample size: 5184; PC=89.5%; EBD=0.0%; POD1=91.7%; POD2=83.2%; POD3=93.5%				

Table 35: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	618	54	0
	$2.8 < \tau < 4.2$	56	561	44
	$\tau > 4.2$	0	59	630
Sample size: 5184; PC=89.5%; EBD=0.0%; POD1=91.7%; POD2=83.2%; POD3=93.5%				

Table 36: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	618	54	0
	$2.8 < \tau < 4.2$	56	561	44
	$\tau > 4.2$	0	59	630
Sample size: 5184; PC=89.5%; EBD=0.0%; POD1=91.7%; POD2=83.2%; POD3=93.5%				

Table 37: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	618	54	0
	$2.8 < \tau < 4.2$	56	561	44
	$\tau > 4.2$	0	59	630
Sample size: 5184; PC=89.5%; EBD=0.0%; POD1=91.7%; POD2=83.2%; POD3=93.5%				

Table 38: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	588	5	28
	$2.8 < \tau < 4.2$	21	611	28
	$\tau > 4.2$	65	58	646
Sample size: 5184; PC=91.2%; EBD=4.6%; POD1=87.2%; POD2=90.7%; POD3=92.0%				

Table 39: Contingency table for variable coherence time (ms) in standard configuration and accuracy 1.22

FATE - Monthly Report

coherence time		Observations		
ms		$\tau < 2.4$	$2.4 < \tau < 3.5$	$\tau > 3.5$
Model data	$\tau < 2.4$	425	0	0
	$2.4 < \tau < 3.5$	0	426	0
	$\tau > 3.5$	1	0	426
Sample size: 3024; PC=99.9%; EBD=0.1%; POD1=99.8%; POD2=100.0%; POD3=100.0%				

Table 40: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

coherence time		Observations		
ms		$\tau < 2.4$	$2.4 < \tau < 3.5$	$\tau > 3.5$
Model data	$\tau < 2.4$	425	0	0
	$2.4 < \tau < 3.5$	0	426	0
	$\tau > 3.5$	1	0	426
Sample size: 3024; PC=99.9%; EBD=0.1%; POD1=99.8%; POD2=100.0%; POD3=100.0%				

Table 41: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

coherence time		Observations		
ms		$\tau < 2.4$	$2.4 < \tau < 3.6$	$\tau > 3.6$
Model data	$\tau < 2.4$	353	0	0
	$2.4 < \tau < 3.6$	0	353	0
	$\tau > 3.6$	1	0	353
Sample size: 2484; PC=99.9%; EBD=0.1%; POD1=99.7%; POD2=100.0%; POD3=100.0%				

Table 42: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	674	1	1
	$2.8 < \tau < 4.2$	0	673	1
	$\tau > 4.2$	0	0	673
Sample size: 5184; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=99.9%; POD3=99.7%				

Table 43: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	674	1	1
	$2.8 < \tau < 4.2$	0	673	1
	$\tau > 4.2$	0	0	673
Sample size: 5184; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=99.9%; POD3=99.7%				

Table 44: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	674	1	1
	$2.8 < \tau < 4.2$	0	673	1
	$\tau > 4.2$	0	0	673
Sample size: 5184; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=99.9%; POD3=99.7%				

Table 45: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	674	1	1
	$2.8 < \tau < 4.2$	0	673	1
	$\tau > 4.2$	0	0	673
Sample size: 5184; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=99.9%; POD3=99.7%				

Table 46: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

coherence time		Observations		
ms		$\tau < 2.8$	$2.8 < \tau < 4.2$	$\tau > 4.2$
Model data	$\tau < 2.8$	674	1	1
	$2.8 < \tau < 4.2$	0	673	1
	$\tau > 4.2$	0	0	673
Sample size: 5184; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=99.9%; POD3=99.7%				

Table 47: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless		Observations		
		glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	400	219	181
	0.5 < glf < 0.7	205	352	366
	glf > 0.7	99	133	157
Sample size: 5184; PC=43.0%; EBD=13.3%; POD1=56.8%; POD2=50.0%; POD3=22.3%				

Table 48: Contingency table for variable ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.5	0.5 < glf < 0.6	glf > 0.6
Model data	glf < 0.5	375	55	3
	0.5 < glf < 0.6	55	309	69
	glf > 0.6	1	67	359
Sample size: 3024; PC=80.7%; EBD=0.3%; POD1=87.0%; POD2=71.7%; POD3=83.3%				

Table 49: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.5	0.5 < glf < 0.6	glf > 0.6
Model data	glf < 0.5	375	55	3
	0.5 < glf < 0.6	55	309	69
	glf > 0.6	1	67	359
Sample size: 3024; PC=80.7%; EBD=0.3%; POD1=87.0%; POD2=71.7%; POD3=83.3%				

Table 50: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.4	0.4 < glf < 0.6	glf > 0.6
Model data	glf < 0.4	293	47	4
	0.4 < glf < 0.6	60	257	51
	glf > 0.6	2	50	299
Sample size: 2484; PC=79.9%; EBD=0.6%; POD1=82.5%; POD2=72.6%; POD3=84.5%				

Table 51: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	619	92	2
	0.5 < glf < 0.7	85	526	106
	glf > 0.7	0	86	596
Sample size: 5184; PC=82.4%; EBD=0.1%; POD1=87.9%; POD2=74.7%; POD3=84.7%				

Table 52: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	619	92
	0.5 < glf < 0.7	85	526
	glf > 0.7	0	86
Sample size: 5184; PC=82.4%; EBD=0.1%; POD1=87.9%; POD2=74.7%; POD3=84.7%			

Table 53: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	619	92
	0.5 < glf < 0.7	85	526
	glf > 0.7	0	86
Sample size: 5184; PC=82.4%; EBD=0.1%; POD1=87.9%; POD2=74.7%; POD3=84.7%			

Table 54: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	619	92
	0.5 < glf < 0.7	85	526
	glf > 0.7	0	86
Sample size: 5184; PC=82.4%; EBD=0.1%; POD1=87.9%; POD2=74.7%; POD3=84.7%			

Table 55: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	619	92
	0.5 < glf < 0.7	85	526
	glf > 0.7	0	86
Sample size: 5184; PC=82.4%; EBD=0.1%; POD1=87.9%; POD2=74.7%; POD3=84.7%			

Table 56: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	589	71
	0.5 < glf < 0.7	16	609
	glf > 0.7	99	24
Sample size: 5184; PC=80.2%; EBD=14.5%; POD1=83.7%; POD2=86.5%; POD3=56.5%			

Table 57: Contingency table for variable ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.6	glf > 0.6
Model data	glf < 0.5	427	3
	0.5 < glf < 0.6	3	427
	glf > 0.6	1	1
Sample size: 3024; PC=99.0%; EBD=0.5%; POD1=99.1%; POD2=99.1%; POD3=98.2%			

Table 58: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.6	glf > 0.6
Model data	glf < 0.5	427	3
	0.5 < glf < 0.6	3	427
	glf > 0.6	1	1
Sample size: 3024; PC=99.0%; EBD=0.5%; POD1=99.1%; POD2=99.1%; POD3=98.2%			

Table 59: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.4	0.4 < glf < 0.6	glf > 0.6
Model data	glf < 0.4	353	0
	0.4 < glf < 0.6	0	353
	glf > 0.6	2	1
Sample size: 2484; PC=99.1%; EBD=0.8%; POD1=99.4%; POD2=99.7%; POD3=96.1%			

Table 60: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	702	6
	0.5 < glf < 0.7	2	698
	glf > 0.7	0	0
Sample size: 5184; PC=99.4%; EBD=0.2%; POD1=99.7%; POD2=99.1%; POD3=98.6%			

Table 61: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	702	6
	0.5 < glf < 0.7	2	698
	glf > 0.7	0	0
Sample size: 5184; PC=99.4%; EBD=0.2%; POD1=99.7%; POD2=99.1%; POD3=98.6%			

Table 62: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

FATE - Monthly Report

ground layer fraction dimensionless		Observations		
		glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	702	6	5
	0.5 < glf < 0.7	2	698	5
	glf > 0.7	0	0	699
Sample size: 5184; PC=99.4%; EBD=0.2%; POD1=99.7%; POD2=99.1%; POD3=98.6%				

Table 63: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

ground layer fraction dimensionless		Observations		
		glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	702	6	5
	0.5 < glf < 0.7	2	698	5
	glf > 0.7	0	0	699
Sample size: 5184; PC=99.4%; EBD=0.2%; POD1=99.7%; POD2=99.1%; POD3=98.6%				

Table 64: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

ground layer fraction dimensionless		Observations		
		glf < 0.5	0.5 < glf < 0.7	glf > 0.7
Model data	glf < 0.5	702	6	5
	0.5 < glf < 0.7	2	698	5
	glf > 0.7	0	0	699
Sample size: 5184; PC=99.4%; EBD=0.2%; POD1=99.7%; POD2=99.1%; POD3=98.6%				

Table 65: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

1.4 Title for PODs tables of Night 1

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PARAMETER	STANDARD	WITH AR (1H)
POD1	67.3	96.0
POD2	73.6	91.9
POD3	76.0	95.0
PC	72.3	94.3
EBD	0.0	0.0

Table 66: PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	93.8	99.2
POD2	46.3	84.7
POD3	87.6	97.9
POD4	44.4	97.0
PC	86.5	97.7
EBD	1.1	0.0

Table 67: PODs for wind direction (degree)

FATE - Monthly Report

PARAMETER	STANDARD	WITH AR (1H)
POD1	56.4	94.1
POD2	48.8	92.4
POD3	76.1	95.3
PC	60.4	93.9
EBD	7.2	0.2

Table 68: PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	95.9	98.8
POD2	71.2	96.5
POD3	72.4	98.2
PC	79.8	97.8
EBD	0.0	0.0

Table 69: PODs for precipitable water vapor (mm)

PARAMETER	STANDARD	WITH AR (1H)
POD1	68.8	91.1
POD2	44.1	82.8
POD3	39.3	91.3
PC	50.8	88.4
EBD	7.4	0.0

Table 70: PODs for total seeing (arcsec) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	92.8	100.0
POD2	96.8	99.2
POD3	69.9	99.5
PC	90.7	99.6
EBD	8.1	0.1

Table 71: PODs for total seeing (arcsec) and accuray 0.24

PARAMETER	STANDARD	WITH AR (1H)
POD1	62.3	91.7
POD2	52.4	83.2
POD3	72.4	93.5
PC	62.4	89.5
EBD	3.8	0.0

Table 72: PODs for coherence time (ms) and accuray 0.0

FATE - Monthly Report

PARAMETER	STANDARD	WITH AR (1H)
POD1	87.2	100.0
POD2	90.7	99.9
POD3	92.0	99.7
PC	91.2	99.9
EBD	4.6	0.0

Table 73: PODs for coherence time (ms) and accuray 1.22

PARAMETER	STANDARD	WITH AR (1H)
POD1	56.8	87.9
POD2	50.0	74.7
POD3	22.3	84.7
PC	43.0	82.4
EBD	13.3	0.1

Table 74: PODs for ground layer fraction (dimensionless) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	83.7	99.7
POD2	86.5	99.1
POD3	56.5	98.6
PC	80.2	99.4
EBD	14.5	0.2

Table 75: PODs for ground layer fraction (dimensionless) and accuray 0.14

2 Title for Day 1

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2.1 Title for Figures of Day 1

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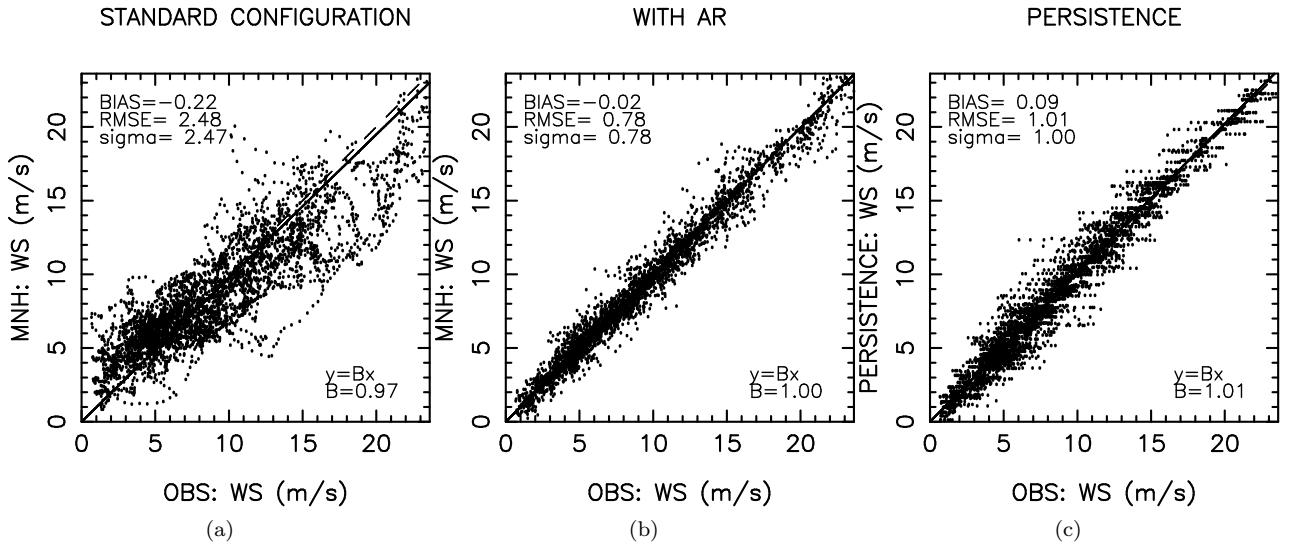


Figure 8: wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

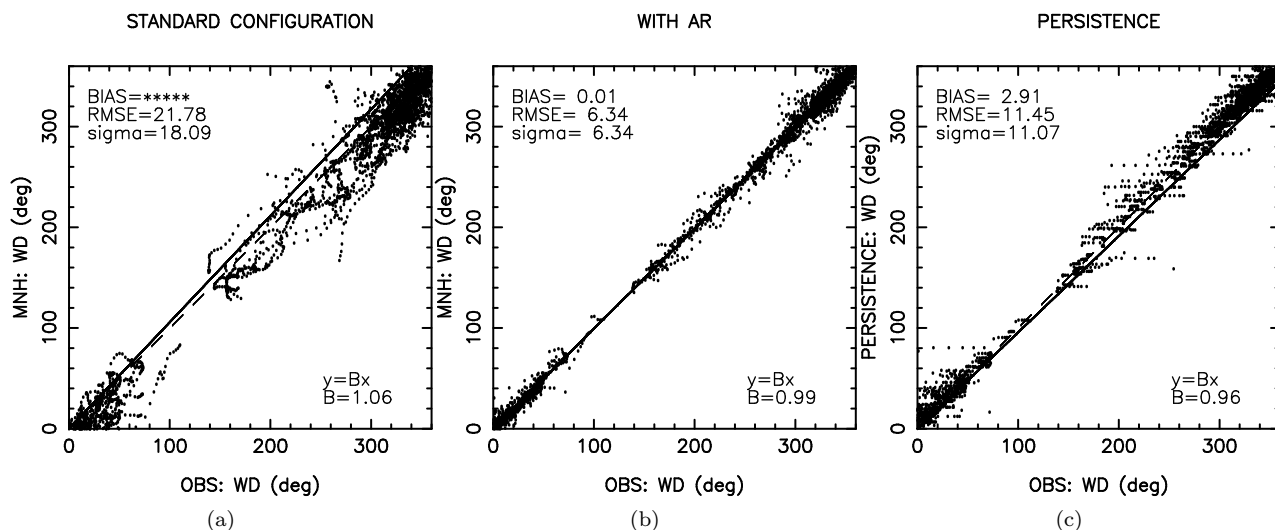


Figure 9: wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

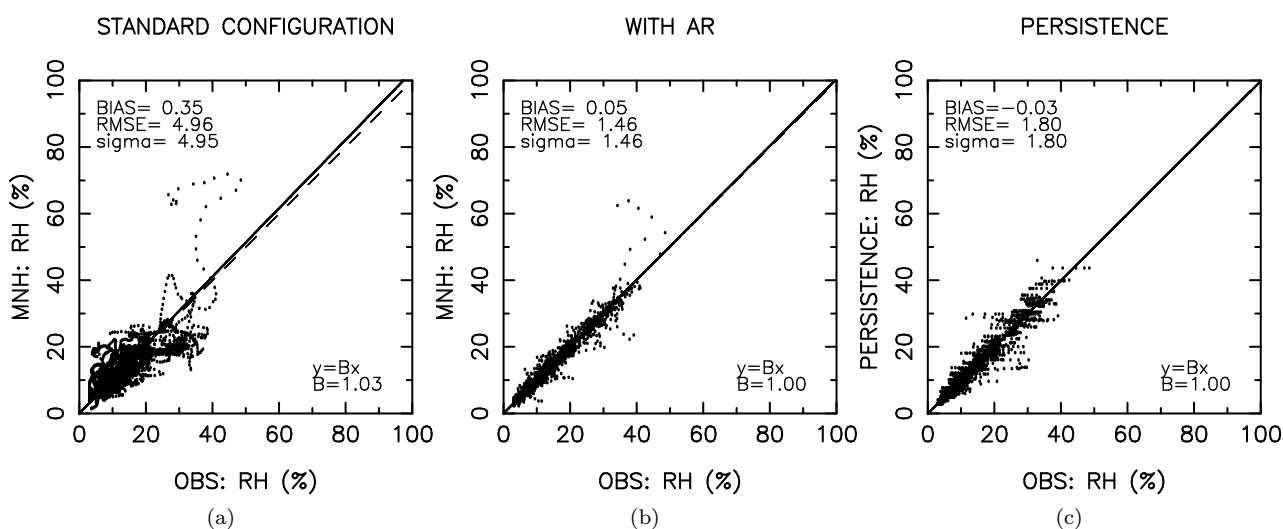


Figure 10: relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

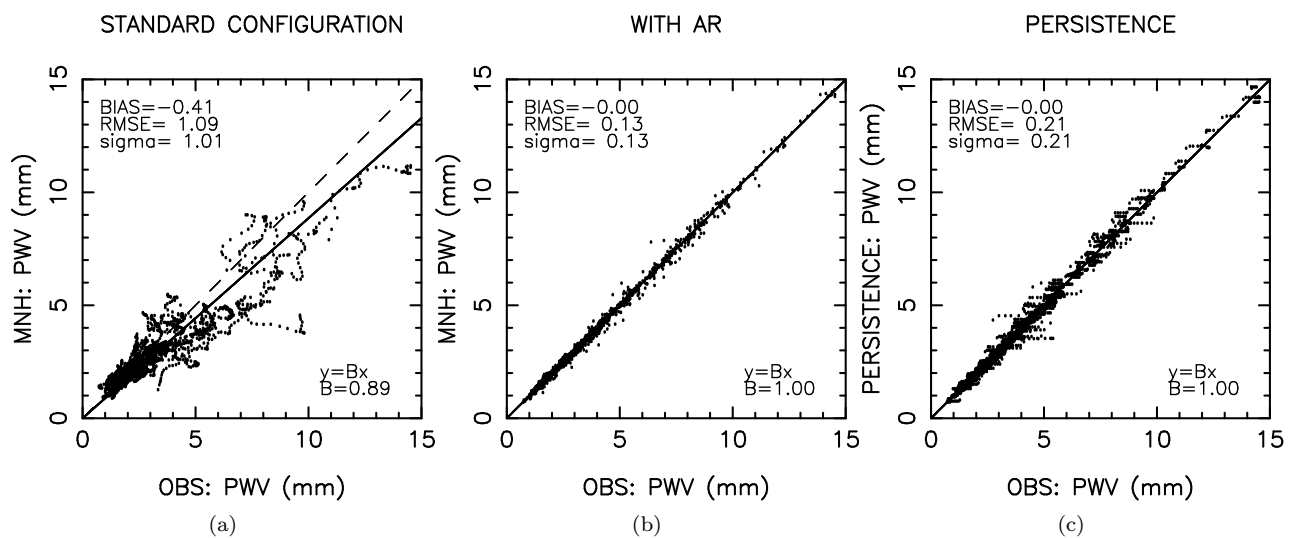


Figure 11: precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H).

2.2 Title for Statistics of Day 1

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VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS ($\text{m}\cdot\text{s}^{-1}$)	-0.22	2.48	2.47	-0.69	2.59	2.50
WD (degree)	-12.13	21.78	18.09	-12.93	20.91	16.44
RH (percent)	0.35	4.96	4.95	0.07	5.49	5.49
PWV (mm)	-0.41	1.09	1.01	-0.62	1.37	1.23
SEE (arcsec)	-	-	-	-	-	-
TAU (ms)	-	-	-	-	-	-
GLF (dimensionless)	-	-	-	-	-	-

Table 76: Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (October-2023)

2.3 Title for Contingency tables of Day 1

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wind speed ms^{-1}	Observations		
	ws < 5.6	5.6 < ws < 10.3	ws > 10.3
Model data	ws < 5.6	710	150
	5.6 < ws < 10.3	626	1072
	ws > 10.3	0	113
Sample size: 6588; PC=70.0%; EBD=0.4%; POD1=53.1%; POD2=80.3%; POD3=76.5%			

Table 77: Contingency table for wind speed (ms^{-1}) in standard configuration

wind direction degree	Observations			
	North	East	South	West
Model data	North	1922	96	0
	East	23	77	8
	South	0	0	254
	West	582	0	10
Sample size: 6588; PC=76.8%; EBD=0.0%; POD1=76.1%; POD2=44.5%; POD3=93.4%; POD4=80.7%				

Table 78: Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 9.0	9.0 < rh < 13.4	rh > 13.4
Model data	rh < 9.0	787	311
	9.0 < rh < 13.4	425	704
	rh > 13.4	115	312
Sample size: 6588; PC=64.6%; EBD=3.1%; POD1=59.3%; POD2=53.1%; POD3=81.4%			

Table 79: Contingency table for variable relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.2	2.2 < pwv < 3.5	pwv > 3.5
Model data	pwv < 2.2	587	121
	2.2 < pwv < 3.5	66	461
	pwv > 3.5	0	71
Sample size: 3294; PC=78.5%; EBD=0.1%; POD1=89.9%; POD2=70.6%; POD3=75.0%			

Table 80: Contingency table for variable precipitable water vapor (mm) in standard configuration

2.4 Title for PODs tables of Day 1

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PARAMETER	STANDARD	WITH AR (1H)
POD1	53.1	91.8
POD2	80.3	90.1
POD3	76.5	94.7
PC	70.0	92.2
EBD	0.4	0.0

Table 81: PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	76.1	96.5
POD2	44.5	85.5
POD3	93.4	97.4
POD4	80.7	91.6
PC	76.8	95.1
EBD	0.0	0.0

Table 82: PODs for wind direction (degree)

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PARAMETER	STANDARD	WITH AR (1H)
POD1	59.3	92.4
POD2	53.1	89.6
POD3	81.4	95.2
PC	64.6	92.4
EBD	3.1	0.0

Table 83: PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	89.9	98.5
POD2	70.6	97.5
POD3	75.0	98.5
PC	78.5	98.2
EBD	0.1	0.0

Table 84: PODs for precipitable water vapor (mm)